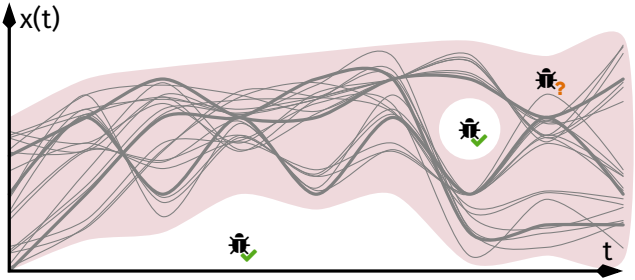

Abstract Interpretation

Software Quality Assurance — Static Code Analysis, II | Florian Sihler | December 10, 2025

Software Engineering
Programming Languages

universität ulm

1. The Why

Why am I even here?

Embedding a Landscape

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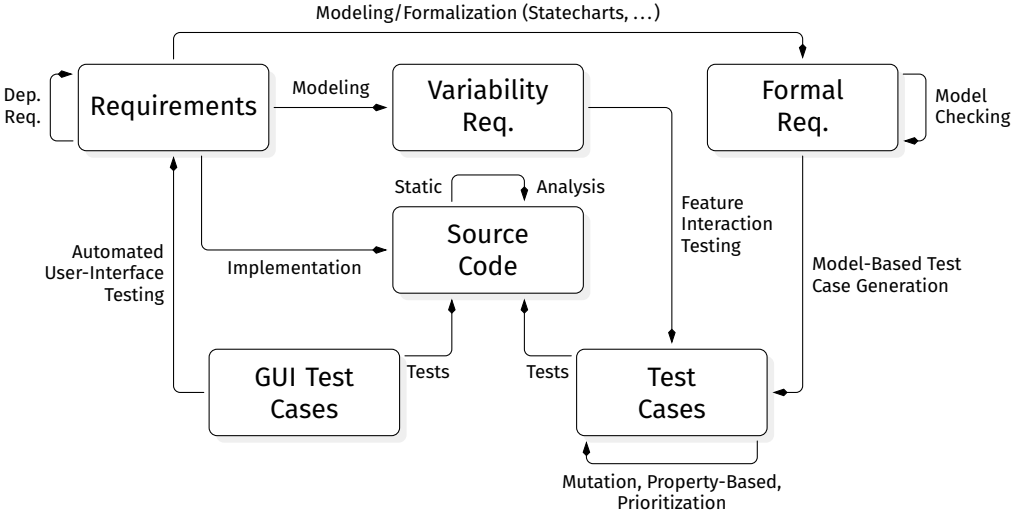

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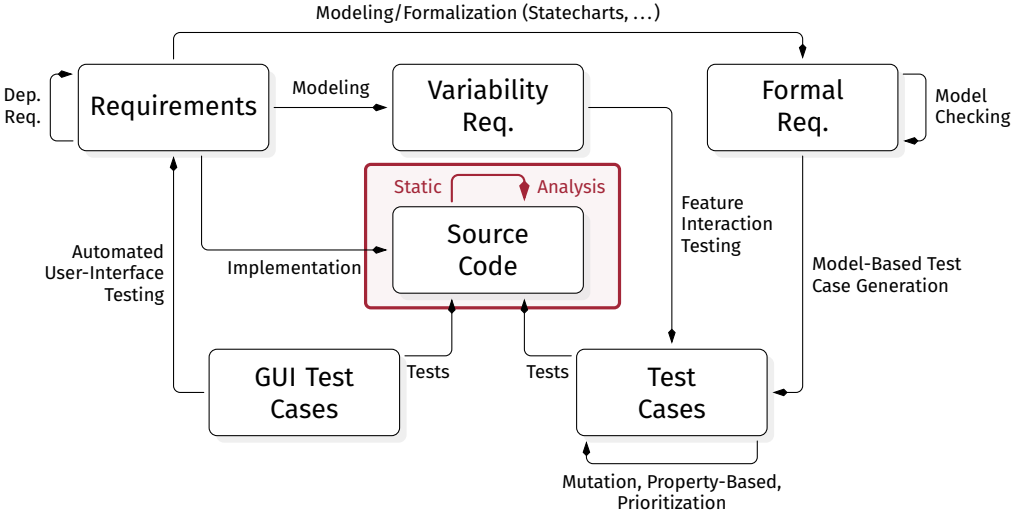

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Today, we learn how to…

- describe semantic properties
- compare, refine, and combine properties
- describe and map language semantics to properties
- deal with the cost of abstraction (and the fun)

The Why

```java
public static void main(String[] args) {
   int a = 1;
   double r = Math.random() * 10;
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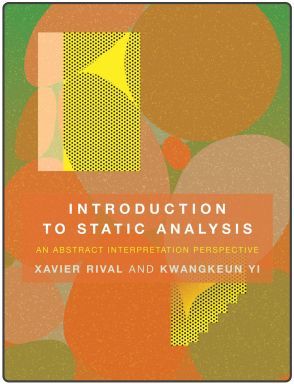
- We want to proof, that a program satisfies certain properties
- Abstract Interpretation is one (/the) technique to do so

Recommended Resources

And for an overview: “Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation” [Min17]

Recommended Resources

Using Analyses [RY20]



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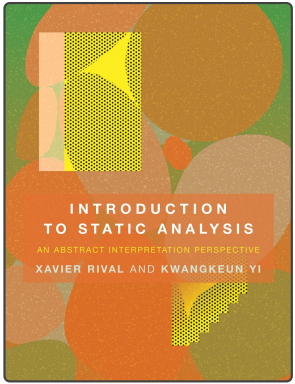

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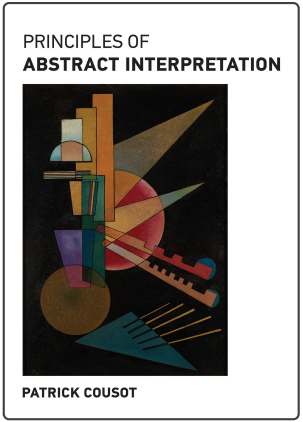

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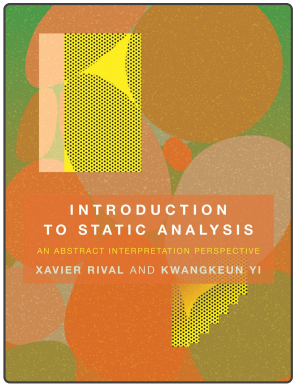

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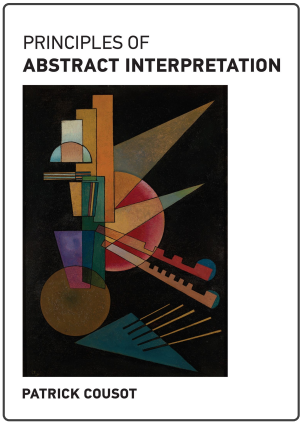
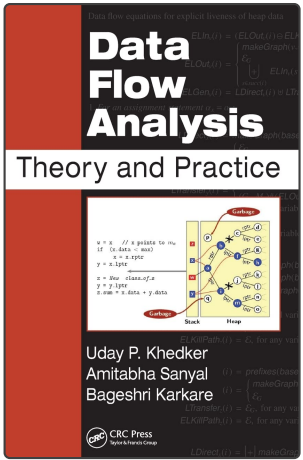


Dataflow Perspective [KSK09]



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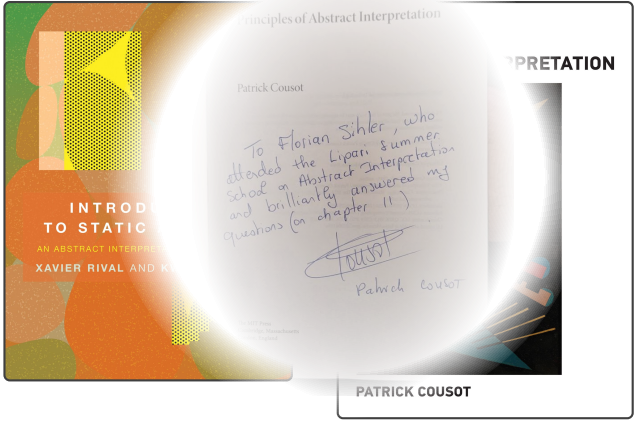

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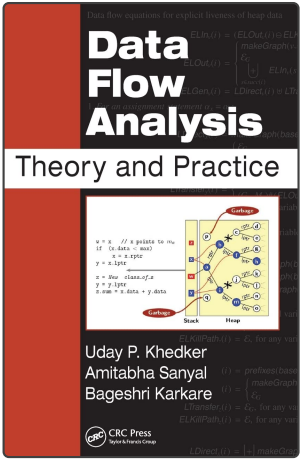

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2. The How

Gimme properties, gimme abstractions!

Abstract Interpretation

See “A casual introduction to Abstract Interpretation” [Cou12]

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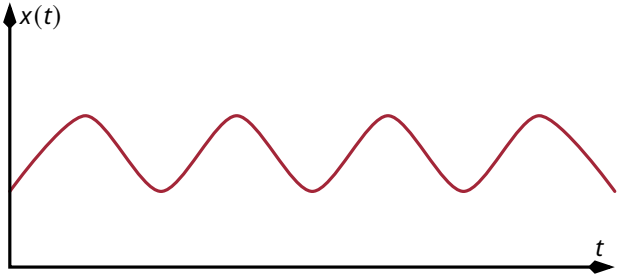

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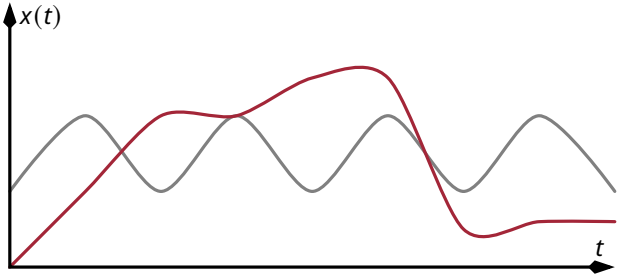

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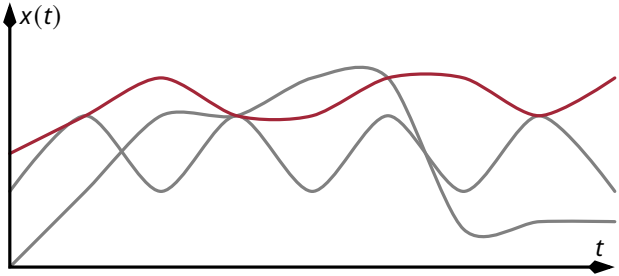

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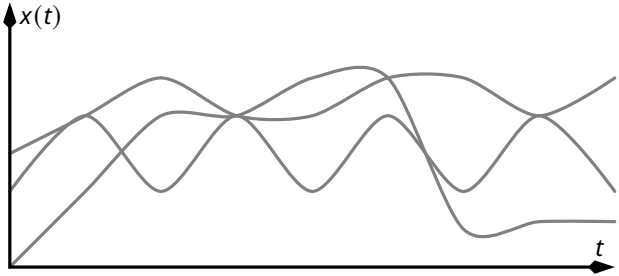

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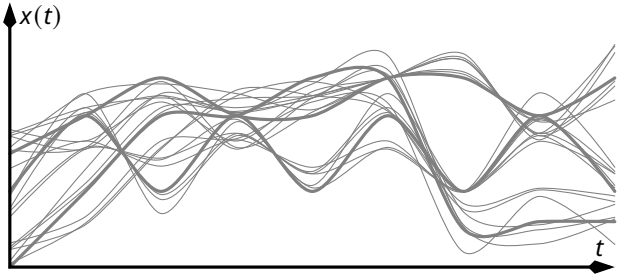

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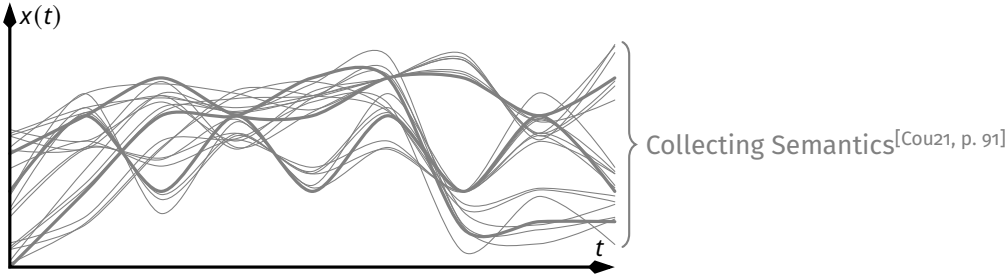

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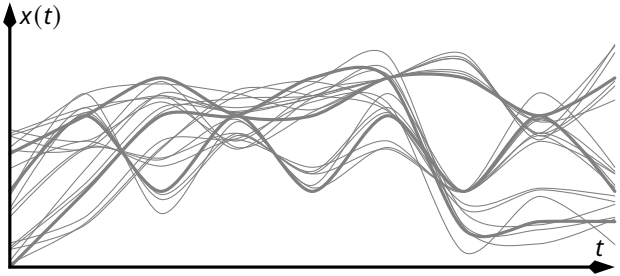

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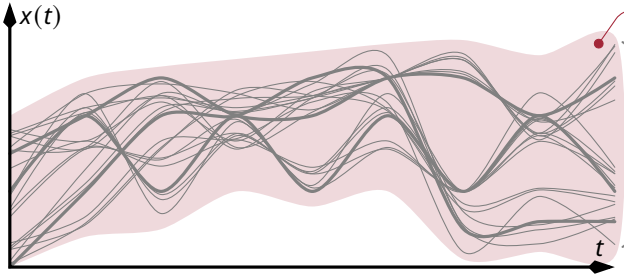

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just one of many
this **must** include all concrete traces!
(over-approximate)

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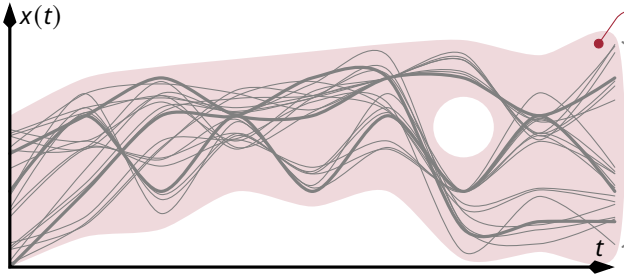

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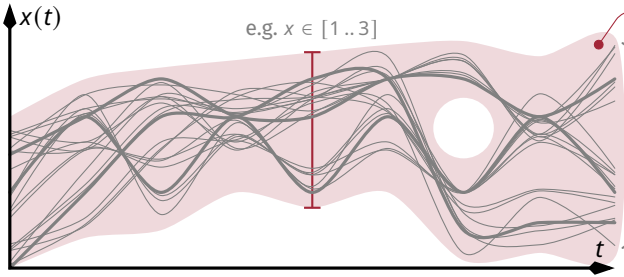

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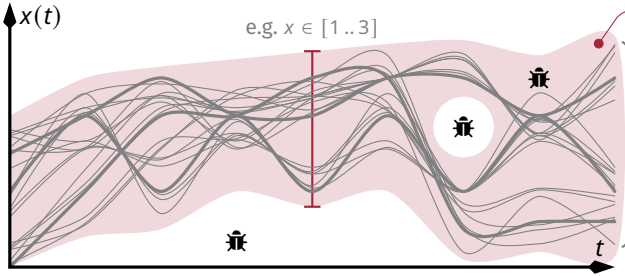

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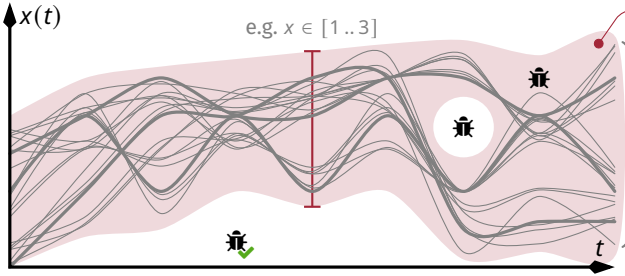

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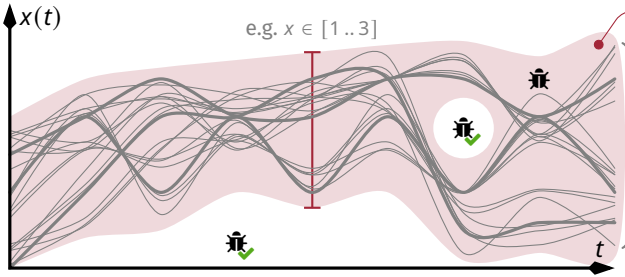

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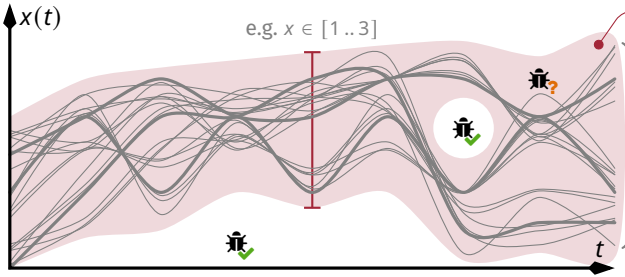

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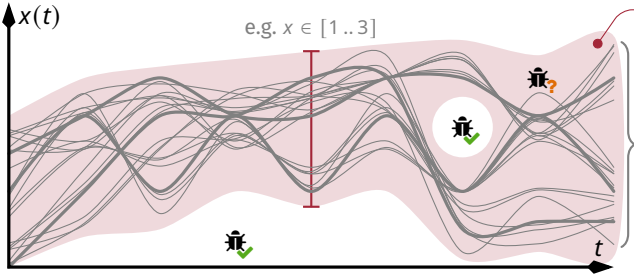
Abstract Interpretation

Radhia Cousot (1947–2014)
Patrick Cousot

Patrick Cousot (1948)
Personal Website



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Terminology

- **Property**

"Principles of Abstract Interpretation" [Cou21, p. 15],"Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 18]

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weaker

weakest

"Principles of Abstract Interpretation" [Cou21, p. 15],"Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 18]

Terminology

- **Property** — Set of states/traces that satisfy that property
  Even integers: $\mathsf{P} = \{ z \in \mathbb{Z} \mid \exists k \in \mathbb{Z} : z = 2k \} = \{ 0, 2, -2, 4, -4, 6, \dots \} \subseteq \mathcal{P}(\mathbb{Z})$ universe ($\mathbb{U}$)

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strongest, stronger, weaker, weakest

- **Partial Order**

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strongest — weakest

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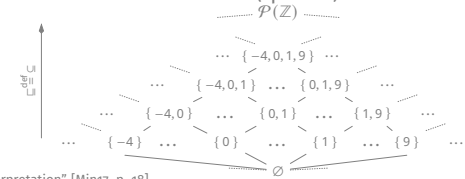
Hasse Diagram of $(\mathcal{P}(\mathbb{Z}), \subseteq)$



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A Task In-Between

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Indicate the greatest and smallest element.

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A directed graph, with edges from a to b ($a \neq b$) indicating that $a \sqsubseteq b$ without a c with $a \sqsubseteq c \sqsubseteq b$.

A Task In-Between

Helmut Hasse (1898–1979)
🅭🅯🄏 2.0 Oberwolfach
Photo Collection

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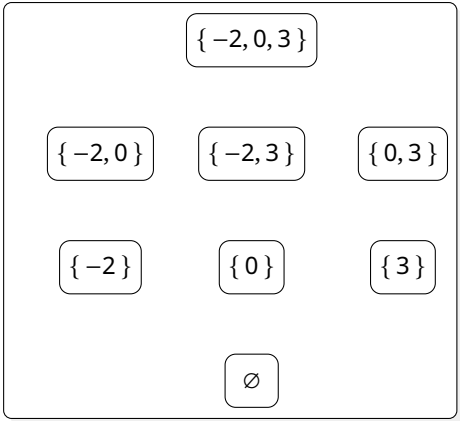

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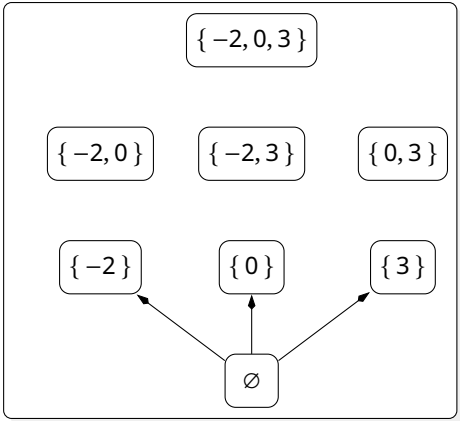

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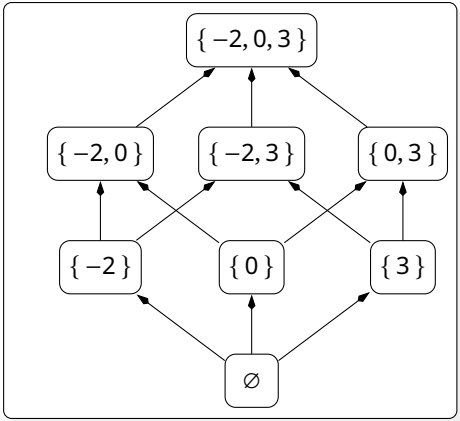

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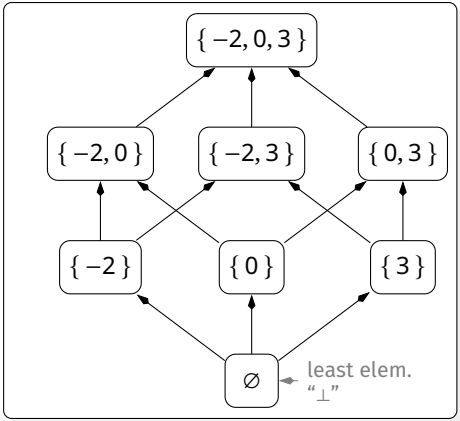

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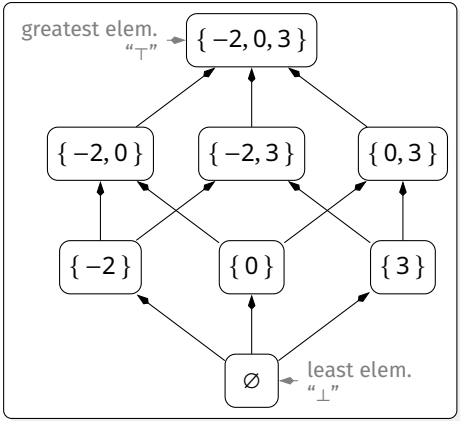

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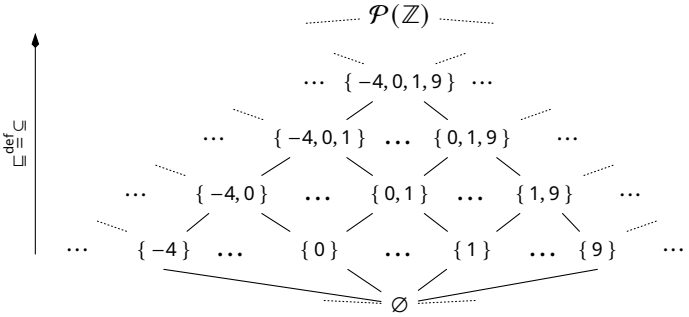
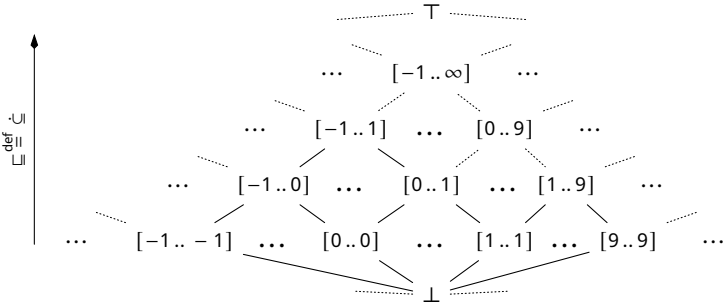
Posets, Lattices, and Chains



"Lattice theory" [Bir67], see also sublattices [Min17, p. 25]

Posets, Lattices, and Chains



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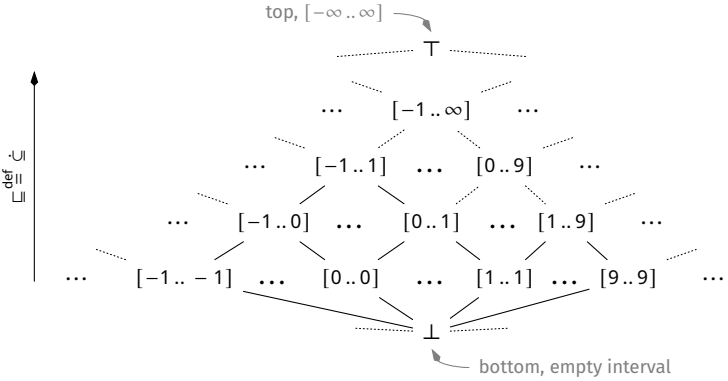

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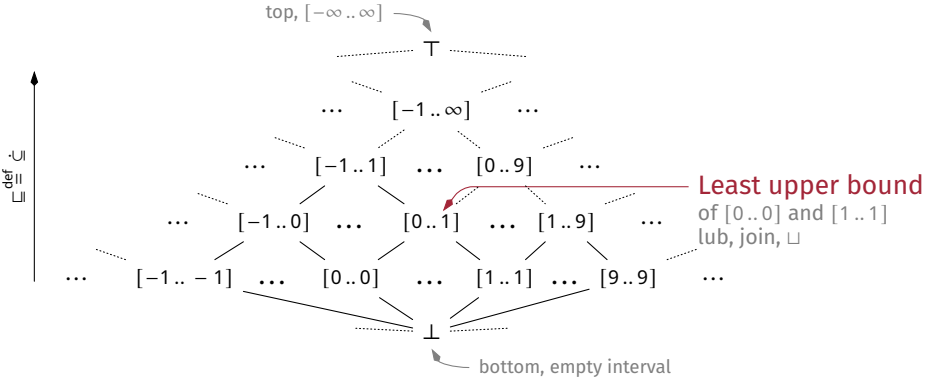

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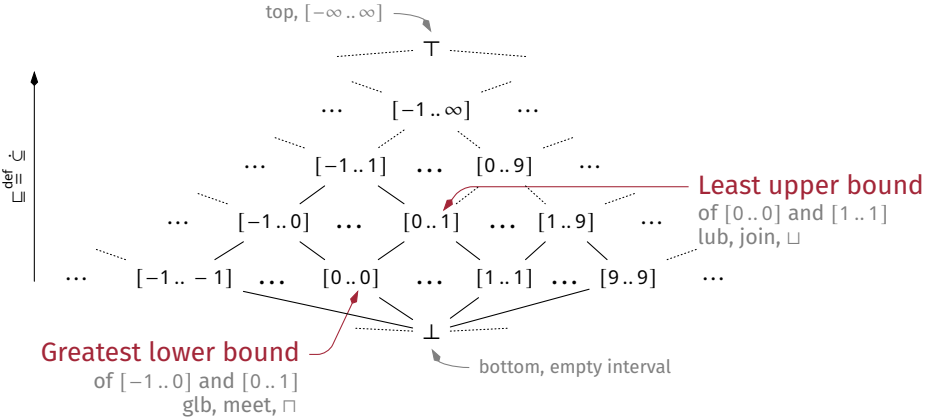

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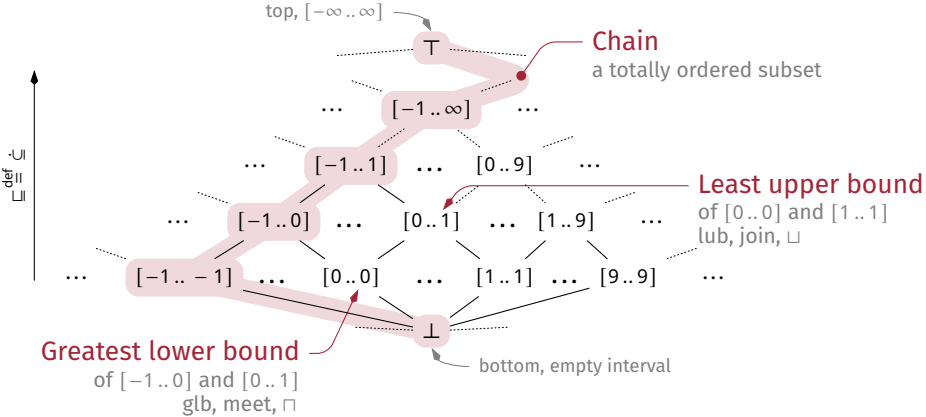

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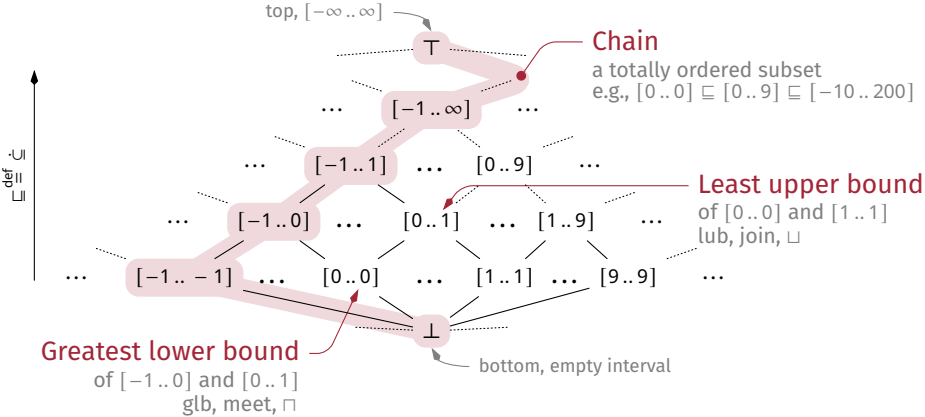

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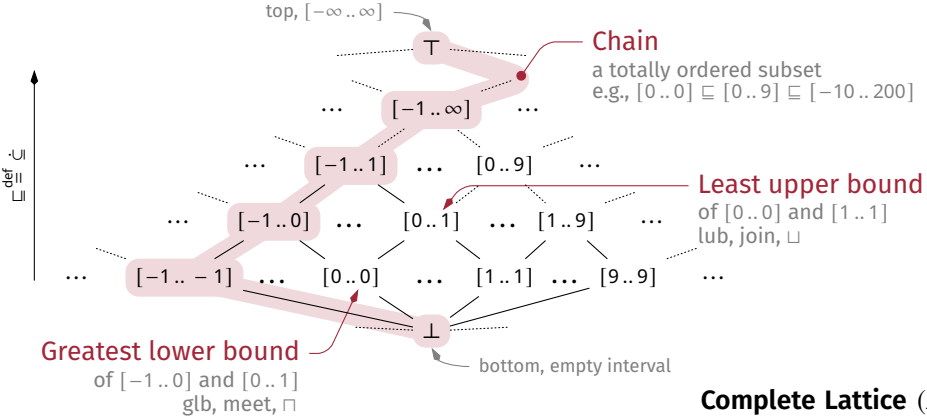

Complete Lattice $(X, \sqsubseteq, \sqcup, \sqcap, \bot, \top)$

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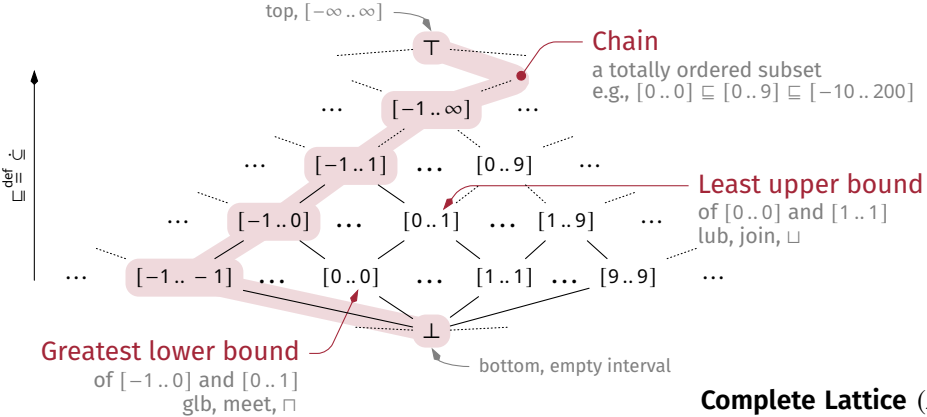

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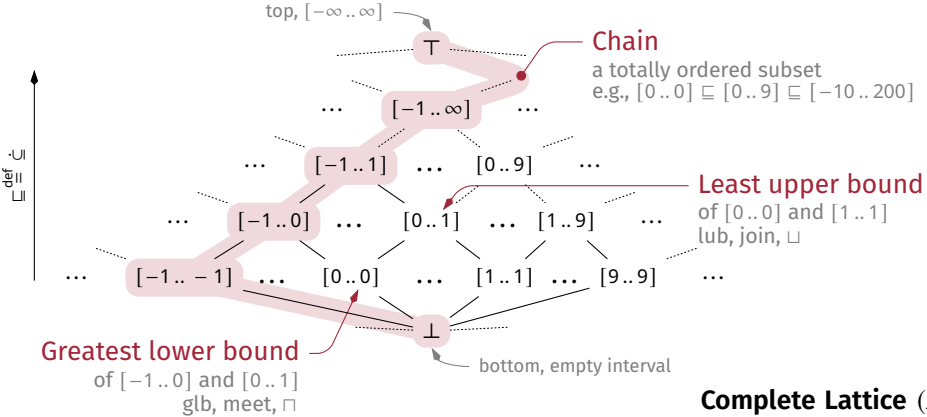

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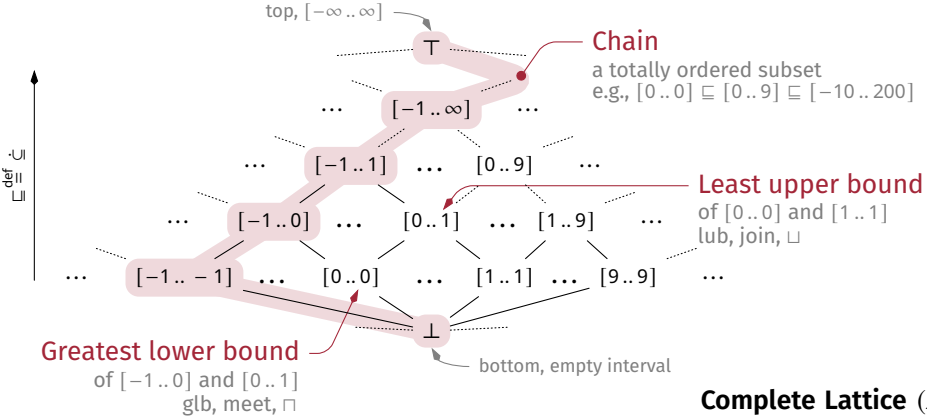

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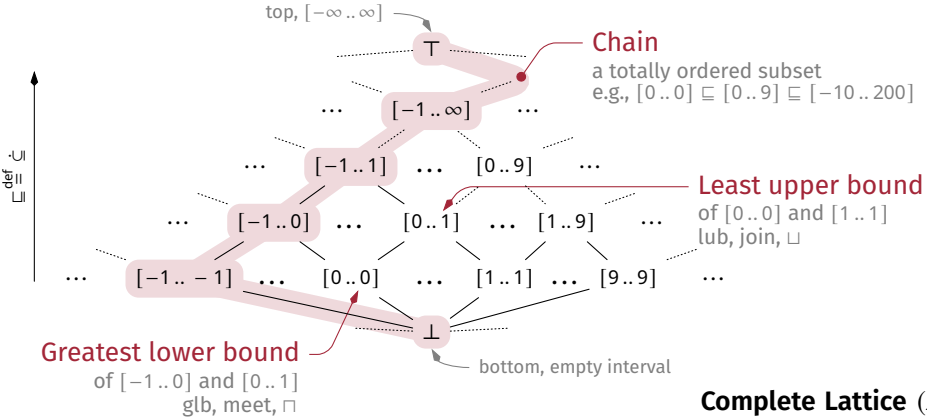

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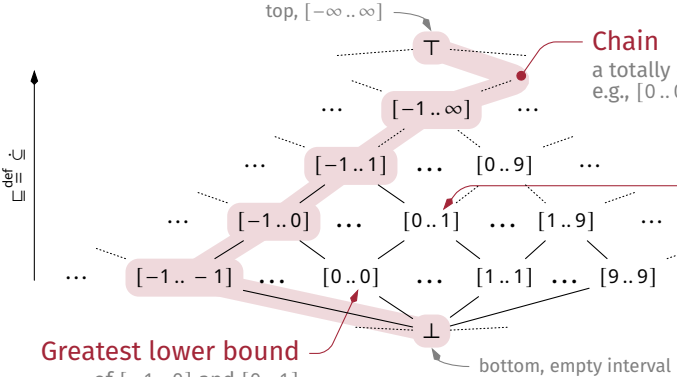
Posets, Lattices, and Chains

Garrett Birkhoff (1911–96)
Paul Halmos



Chain
a totally ordered subset
e.g., $[0\,..\,0] \sqsubseteq [0\,..\,9] \sqsubseteq [-10\,..\,200]$

Least upper bound
of $[0\,..\,0]$ and $[1\,..\,1]$
lub, join, $\sqcup$

Greatest lower bound
of $[-1\,..\,0]$ and $[0\,..\,1]$
glb, meet, $\sqcap$

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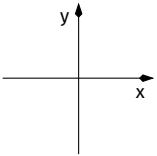
Abstract Domains

Pick Your Favorite Lattice!

Numerical



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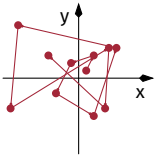

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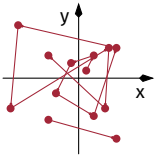

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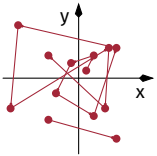

Collecting Semantics

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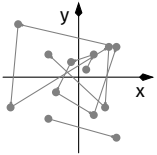

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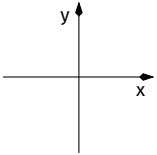

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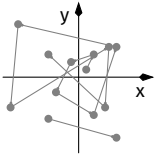

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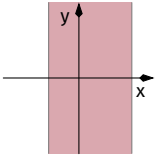
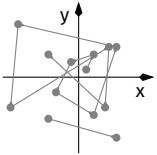


"Principles of Abstract Interpretation" [Cou21], "Pentagons: a weakly relational abstract domain for the efficient validation of array accesses" [LF08, p. 25]

Abstract Domains

Pick Your Favorite Lattice!

Numerical



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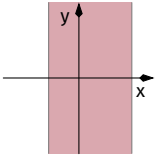

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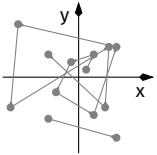

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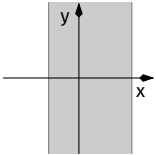

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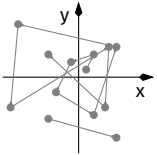

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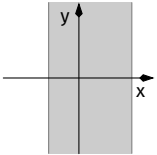

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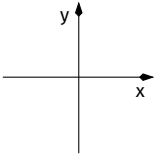

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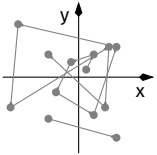

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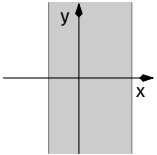

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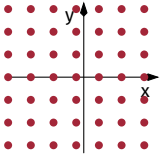

Simple Congruences

"Principles of Abstract Interpretation" [Cou21], "Pentagons: a weakly relational abstract domain for the efficient validation of array accesses" [LF08, p. 25]

Abstract Domains

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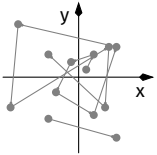

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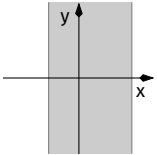

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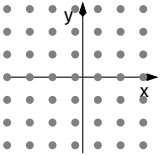

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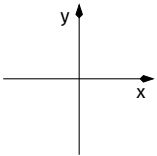

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Abstract Domains

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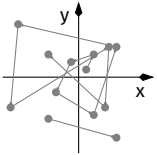

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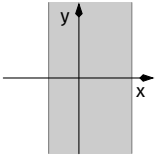

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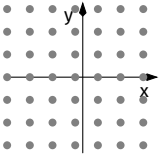

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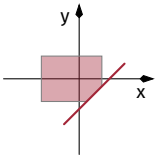

Pentagons

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Abstract Domains

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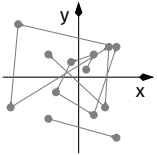

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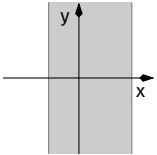

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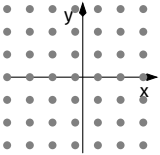

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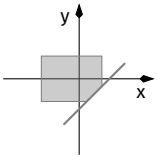

Pentagons

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Abstract Domains

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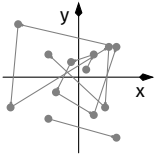

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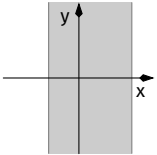

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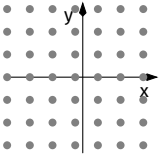

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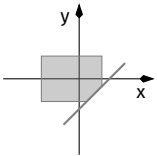


Pentagons

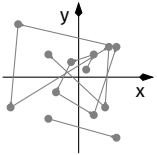
- Octagons

"Principles of Abstract Interpretation" [Cou21], "Pentagons: a weakly relational abstract domain for the efficient validation of array accesses" [LF08, p. 25]

Abstract Domains

Pick Your Favorite Lattice!

Numerical



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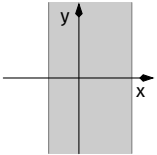

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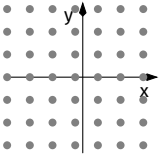

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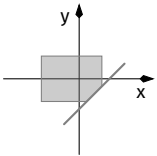

Pentagons

- Octagons
- Ellipses

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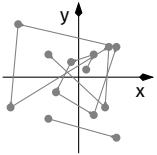

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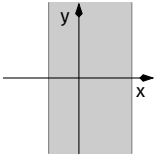

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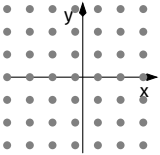

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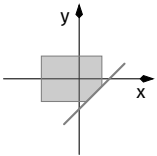

Pentagons

- Octagons
- Ellipses
- Exponentials

"Principles of Abstract Interpretation" [Cou21], "Pentagons: a weakly relational abstract domain for the efficient validation of array accesses" [LF08, p. 25]

Abstract Domains

Pick Your Favorite Lattice!

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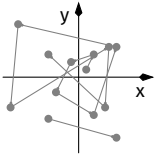

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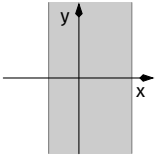

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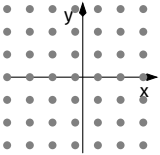

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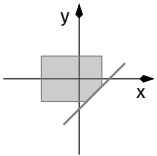


Pentagons

- Octagons
- Ellipses
- Exponentials
- Signs

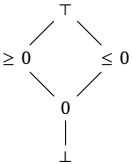
"Principles of Abstract Interpretation" [Cou21], "Pentagons: a weakly relational abstract domain for the efficient validation of array accesses" [LF08, p. 25]

Sign Analysis

Simple Sign Domain

Sign Analysis

Simple Sign Domain



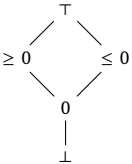
Sign Analysis

Simple Sign Domain

- We still have no program semantics, but we can try…

```
int a = 0;
int b = 12;
int c = a + b;
int d = c - b;
```

java



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$\wr a = 0 \int$

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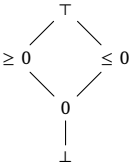

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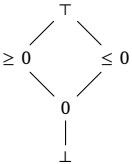
$\wr a = 0 \int$

$\wr b \geq 0 \int$

java



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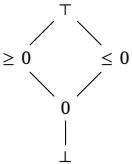
$\wr a = 0 \int$

$\wr b \geq 0 \int$

$\wr c \geq 0 \quad (= 0 + \geq 0) \int$

java



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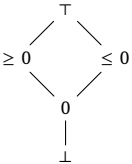
$\wr a = 0 \int$

$\wr b \geq 0 \int$

$\wr c \geq 0 \quad (= 0 \ + \ \geq 0) \int$

$\wr d = \top \quad (\geq 0 \ - \ \geq 0) \int$

java



Sign Analysis

Simple Sign Domain

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```

java

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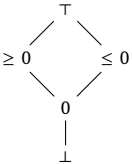

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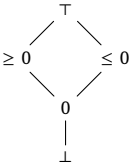

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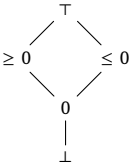

Galois Connection — Linking Posets [Cou21, p. 110]
(small detour)

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The abstract domain we choose, e.g., intervals

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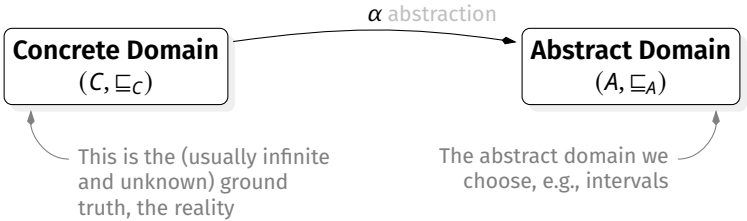

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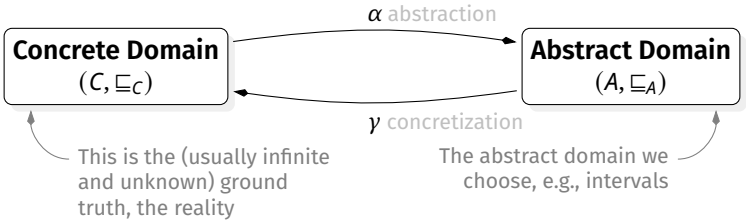

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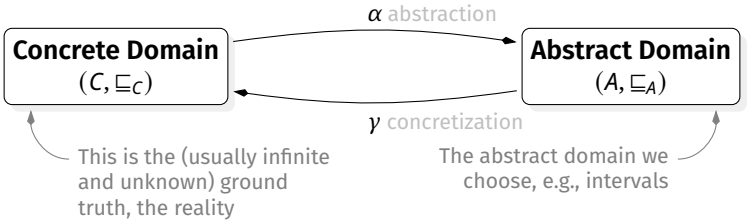

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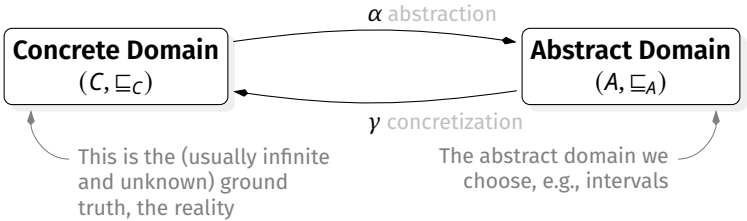

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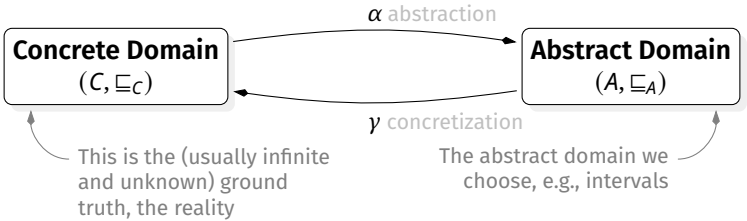

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Évariste Galois (1811–32)
Public Domain
died in a duel for love

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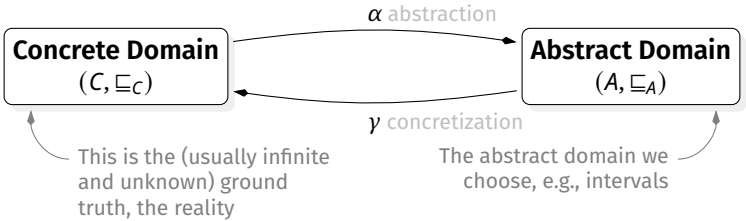

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- But, how do we handle loops? Recursion?

Fixpoints

"A lattice-theoretical fixpoint theorem and its applications." [Tar55], "Introduction to metamathematics" [Kle52], "Principles of Abstract Interpretation" [Cou21, p. 165]

Fixpoints

- For operators $f : X \longrightarrow X$ a **fixpoint** is a $x \in X$ such that $f(x) = x$

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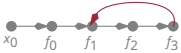
  - 

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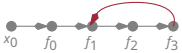
  - 

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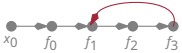
  - 

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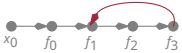
  - 

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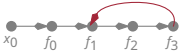
  - 

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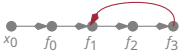
  - 

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for complete, nonempty lattices
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Alfred Tarski (1901–83)
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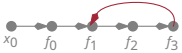
  - 

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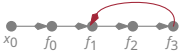
  - 

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- Analyzing, e.g. loops, we "go up" the lattice until we reach a least fixpoint

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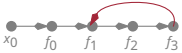
  - 

    reach a cycle, $f^{p+\ell} = f^p, \ell > 0$
  -

    iterate forever, $\forall p \neq q \in \mathbb{N} : f^p \neq f^q$

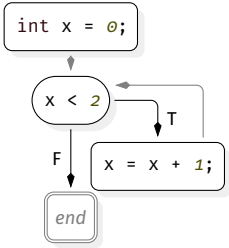
$f : \mathbb{N} \to \mathbb{N}, f(x) = x + 1$

- If our function is monotonic, we can always find a fixpoint[Tar55]

  for complete, nonempty lattices
  Tarski's Theorem
- Analyzing, e.g. loops, we "go up" the lattice until we reach a least fixpoint
- You may know fixpoints from tools like LaTeX or the λ-calculus

"A lattice-theoretical fixpoint theorem and its applications." [Tar55], "Introduction to metamathematics" [Kle52], "Principles of Abstract Interpretation" [Cou21, p. 165]

Interval Analysis, I (the intuitive approach)



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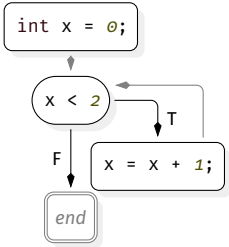

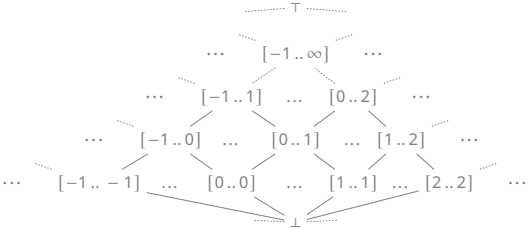

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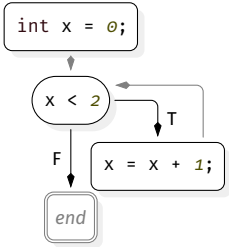
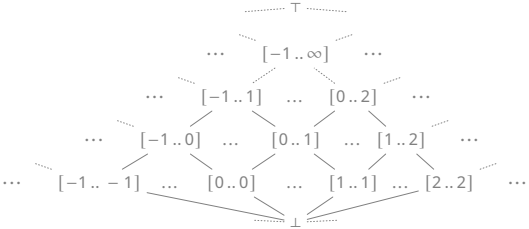


$\wr\, x_0 \in [0\,..\,0] \,\wr$



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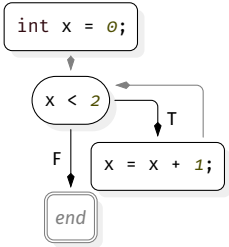

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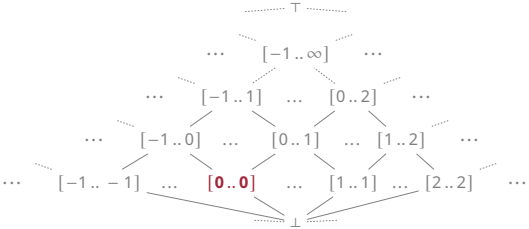

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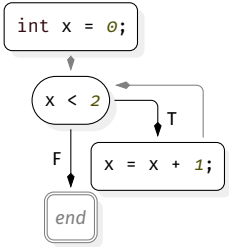

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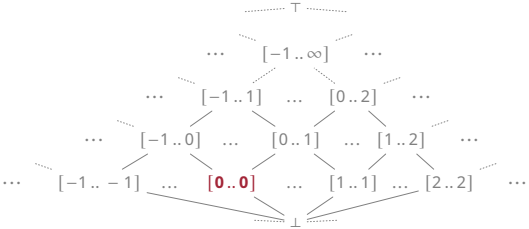

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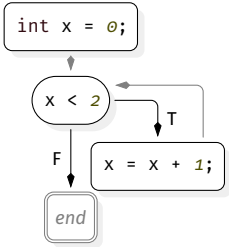

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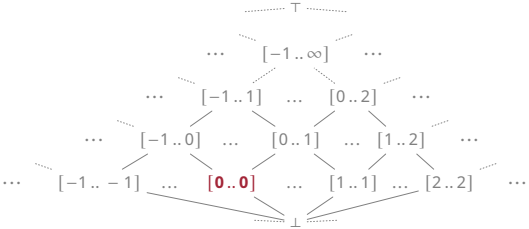
if we are inside the loop we know that $x < 2$ holds!



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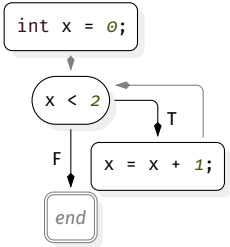

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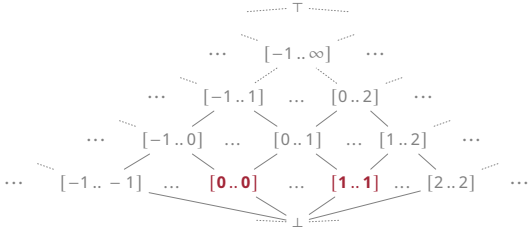
$\wr\, x_3 \in [1\,..\,1] \quad ([0\,..\,0] \oplus [1\,..\,1]) \,\wr$

we have to know how to add intervals here



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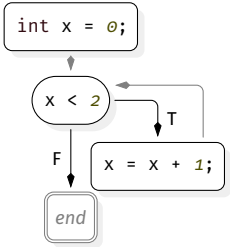

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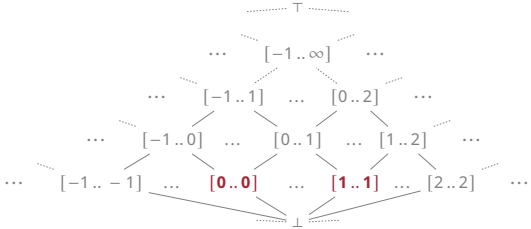

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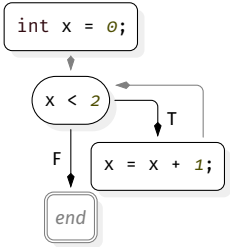

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we join on loops because, at this point x can have any of the values

$\wr$ [pre] $\boldsymbol{x_1 \in [0\,..\,1]}$ $\quad$ $\boldsymbol{([0\,..\,0] \cup [1\,..\,1])}\ \int$

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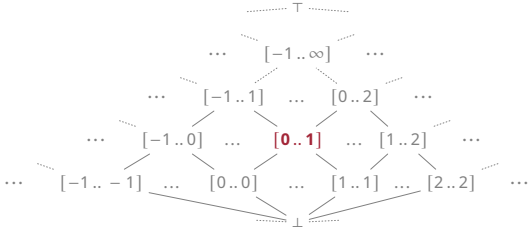

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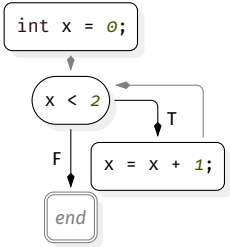

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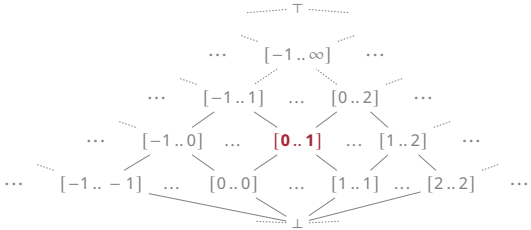

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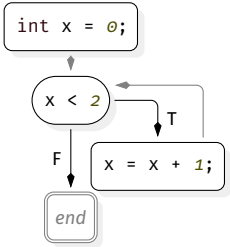

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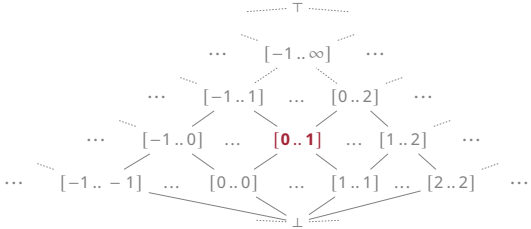

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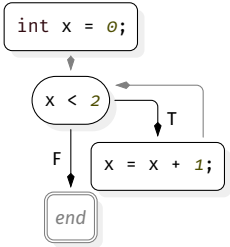

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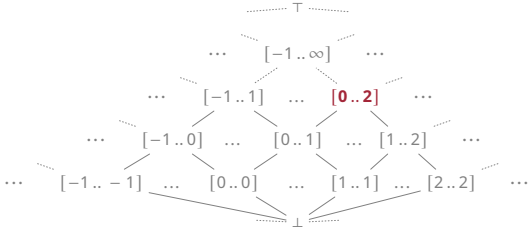

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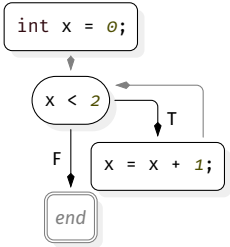

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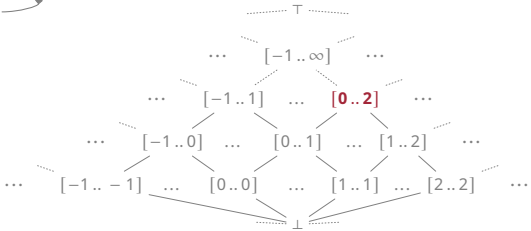
$\wr$ [post] $x_4 \in [2\,..\,2] \quad ([0\,..\,2] \cap [2\,..\,\infty))\ \int$

after the loop we know that $\neg(x < 2) = x \geq 2$ holds!



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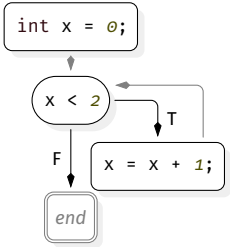

Intervals

$\wr\, x_0 \in [0\,..\,0] \,\wr$

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$\wr$ [post] $x_4 \in [2\,..\,2] \quad ([0\,..\,2] \cap [2\,..\,\infty)) \,\wr$

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Signs

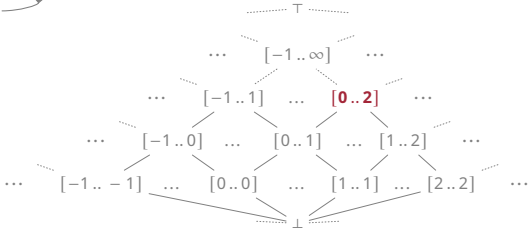
$\wr\, x_0 = 0 \,\wr$

$\wr$ [pre] $x_1 \geq 0 \,\wr$

$\wr$ [in] $x_2 \geq 0 \,\wr$

$\wr\, x_3 \geq 0 \,\wr$

$\wr$ [post] $x_4 \geq 0 \,\wr$



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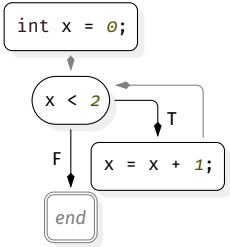

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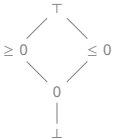

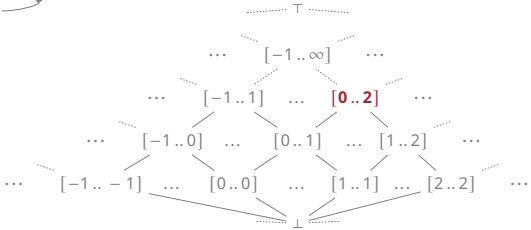

3. Semantics

What does my program mean?

Semantics

Shortened form of “Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation” [Min17, p. 46], craftinginterpreters.com/representing-code.html

Program Syntax (simplified)

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

Program Syntax (simplified)

```java
int x = 0;

while(x < 2) {

  x = x + 1;

}
```

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

Program Syntax (simplified)

Variable $v \in \mathbb{V}$

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Program Syntax (simplified)

Variable $v \in \mathbb{V}$

Assignment

```
int x = 0;
```

Numeric Constant $c \in \mathbb{I}$

```
while(x < 2) {

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```

java

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

Program Syntax (simplified)

Variable $v \in \mathbb{V}$

Assignment

Sequence

Numeric Constant $c \in \mathbb{I}$

```
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java

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

Program Syntax (simplified)

Variable $v \in \mathbb{V}$

Assignment

Sequence

Numeric Constant $c \in \mathbb{I}$

Loop

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Program Syntax (simplified)

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Sequence

Numeric Constant $c \in \mathbb{I}$

Loop

Comparison $\bowtie \in \{\leq, <, \ldots\}$

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Program Syntax (simplified)

Variable $v \in \mathbb{V}$

Assignment

```
int x = 0;
```

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Numeric Constant $c \in \mathbb{I}$

Loop

```
while(x < 2) {
```

Comparison $\bowtie \in \{\leq, <, \ldots\}$

```
  x = x + 1;
```

Binary Expression

```
}
```

java

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Variable $v \in \mathbb{V}$

Assignment

Sequence

Numeric Constant $c \in \mathbb{I}$

Loop

Comparison $\bowtie \in \{\leq, <, \ldots\}$

Binary Expression

stm	::=	$V \leftarrow$ *expr*	(assignment, $V \in \mathbb{V}$)	java
	|	stm_1; stm_2	(sequence)	
	|	**while**(*cond*) { *stm* }	(loop)	
expr	::=	V	(variable, $V \in \mathbb{V}$)	
	|	c	(constant, $c \in \mathbb{I}$)	
	|	$expr_1 \diamond expr_2$	(bin. expr., $\diamond \in \{+, -, \ldots\}$)	
cond	::=	b	(boolean, $b \in \mathbb{B}$)	
	|	$expr_1 \bowtie expr_2$	(comparison, $\bowtie \in \{\leq, <, \ldots\}$)	

Shortened form of "Tutorial on Static Inference of Numeric Invariants by Abstract Interpretation" [Min17, p. 46], craftinginterpreters.com/representing-code.html

Atomic Expression Semantics

```
int x = 0;
while(x < 2) {
   x = x + 1;
}
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java

$expr$	$::=$	V	(variable, $V \in \mathbb{V}$)
	$\mid$	c	(constant, $c \in \mathbb{I}$)
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- We use an environment $\mathcal{E} \stackrel{\text{def}}{=} \mathbb{V} \to \mathbb{I}$ to represent the current program state

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$\mathbb{V}$	$\mathbb{I}$
x	0
c	5

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Integer Values

- We use an environment $\mathcal{E} \overset{\text{def}}{=} \mathbb{V} \to \mathbb{I}$ to represent the current program state
- Now we can define evalExpr(*expr*, *env*) for an environment $env \in \mathcal{E}$

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Integer Values

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Usually written as $\mathbb{E}[\![\, expr \,]\!]\rho$

- Now we can define evalExpr($expr$, env) for an environment $env \in \mathcal{E}$

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Value of $V \in \mathbb{V}$ in Environment env

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Value of $V \in \mathbb{V}$ in Environment env

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$$\text{evalExpr}(expr_1 + expr_2,\ env) \stackrel{\text{def}}{=} \text{evalExpr}(expr_1,\ env) + \text{evalExpr}(expr_2,\ env)$$

$\mathbb{V}$	$\mathbb{I}$
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```

java

$$
\begin{array}{rcll}
\mathit{expr} & ::= & V & \text{(variable, } V \in \mathbb{V}\text{)} \\
 & | & c & \text{(constant, } c \in \mathbb{I}\text{)} \\
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\end{array}
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- Now we can define evalExpr(expr, env) for an environment $\mathit{env} \in \mathcal{E}$

$$
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\text{evalExpr}(V,\ \mathit{env}) &\stackrel{\text{def}}{=} \mathit{env}(V) \\
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&\ \vdots
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Value of $V \in \mathbb{V}$ in Environment env

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x	0
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- Additionally, we can define evalCond(cond, envs) and evalStm(stm, envs)

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Denotational Semantics while loops

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while(*cond*) { *stm* }

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Suppose we start the loop with states *Start*

$$\textbf{while}(\textit{cond})\ \{\ \textit{stm}\ \}$$

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$$\textbf{while}(cond)\ \{\ stm\ \} \qquad F(X) \stackrel{\text{def}}{=} Start \cup evalStm(stm,\ evalCond(cond,\ X))$$

iterate to find the least fixpoint [Min17, p. 52]

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Keep only states S with $evalCond(\neg cond,\ S)$

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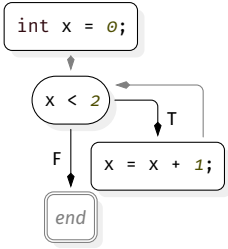
Usually written as $\mathbb{S}^\#, \mathbb{C}^\#, \mathbb{E}^\#, \ldots$

Interval Analysis, II



"Principles of Abstract Interpretation" [Cou21, p. 393], "Widening and narrowing operators for abstract interpretation" [CZ11]

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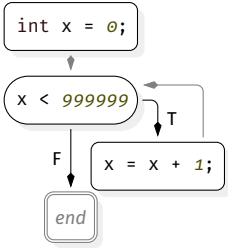

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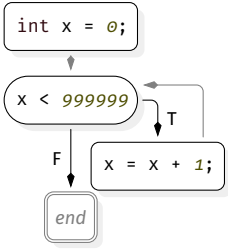

$$\wr\, x_0 \in [0\,..\,0] \,\int$$

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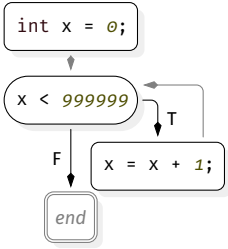

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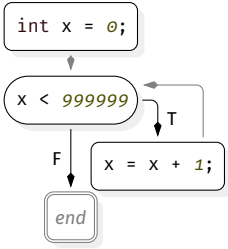

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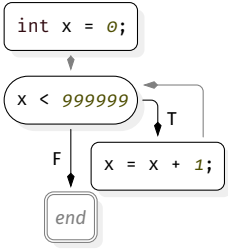

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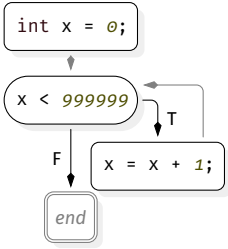

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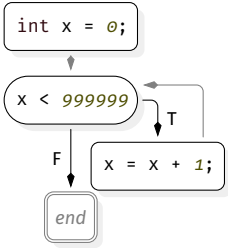

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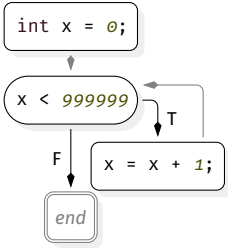

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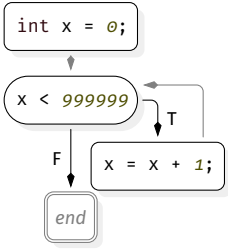

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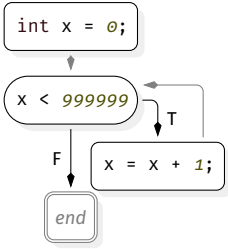

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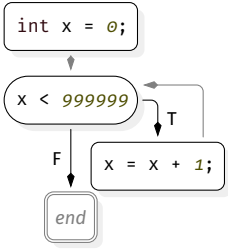

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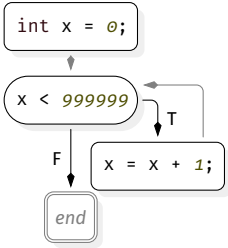

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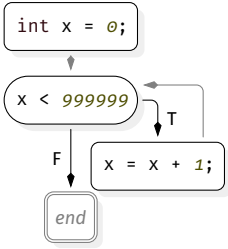

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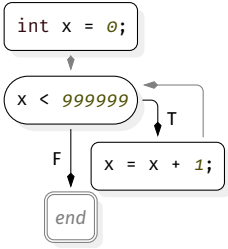

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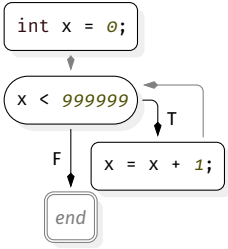

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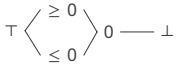

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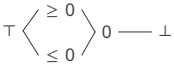

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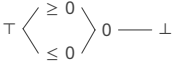

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   We do not need widening here as our lattice is finite and our semantics do not introduce new elements. (We will concretize this next week.)

4. Outlook and Comments

This is incredible, I need more!

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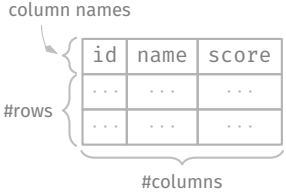

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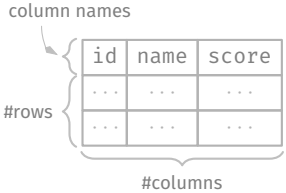

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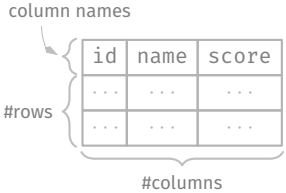

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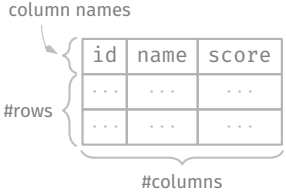

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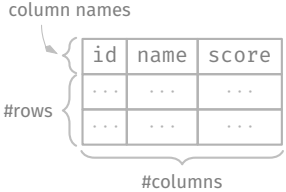

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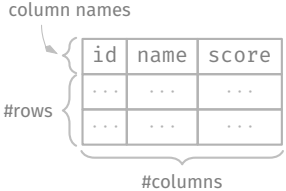

Category theory is amazing!

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column names

id	name	score
...	...	...
...	...	...

#rows

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[Ger25] Oliver Gerstl. Tracking the shape of data frames in R programs using abstract interpretation (Ulm University, 2025)

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Domains can also capture relations between variables (e.g. *polyhedra*), their provenance, and much more!

However, this implies trade-offs which we discuss next time.

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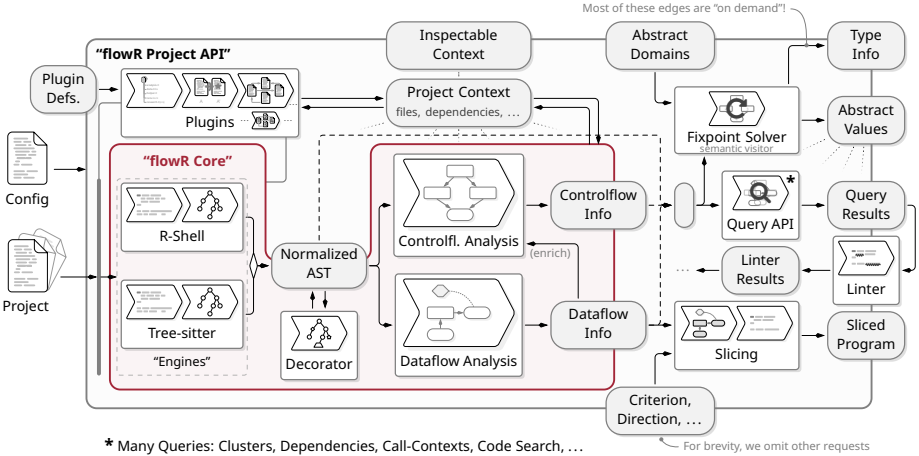

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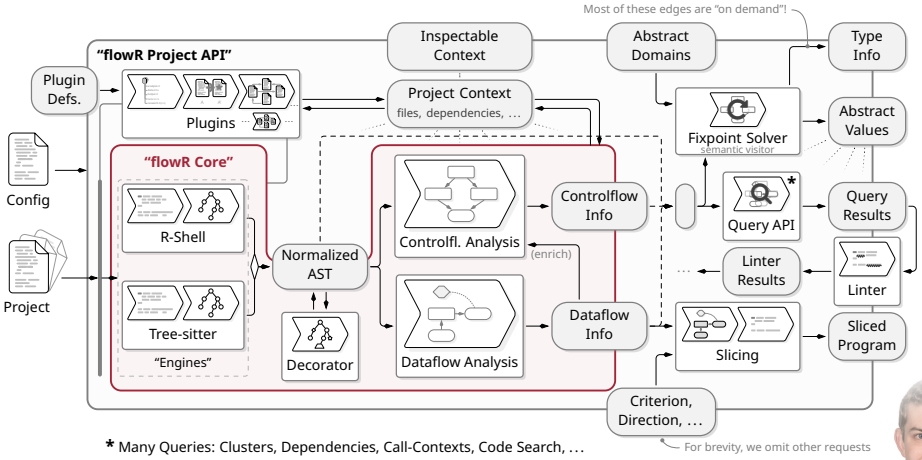

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A Little Bit of History

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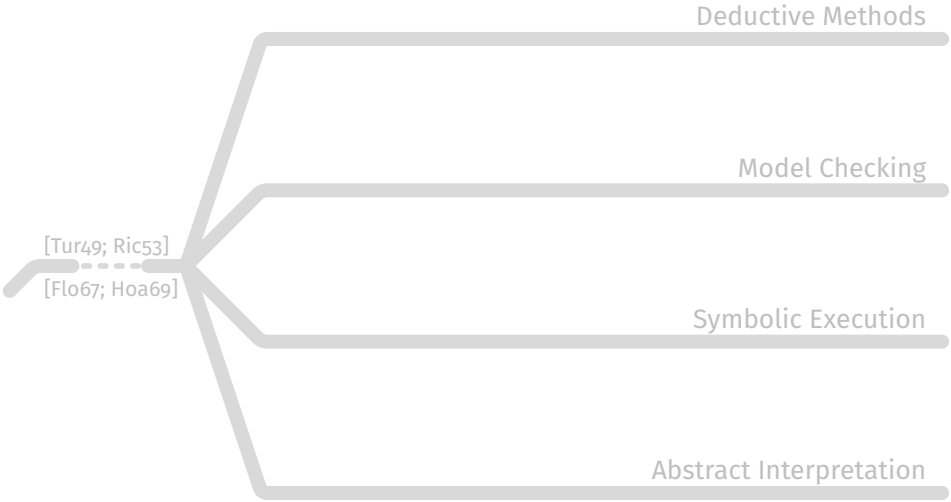

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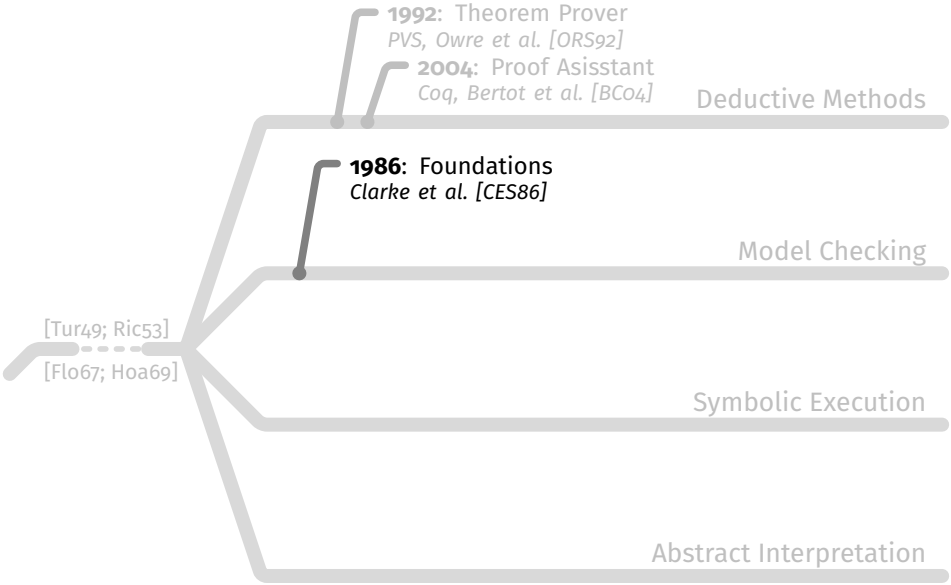

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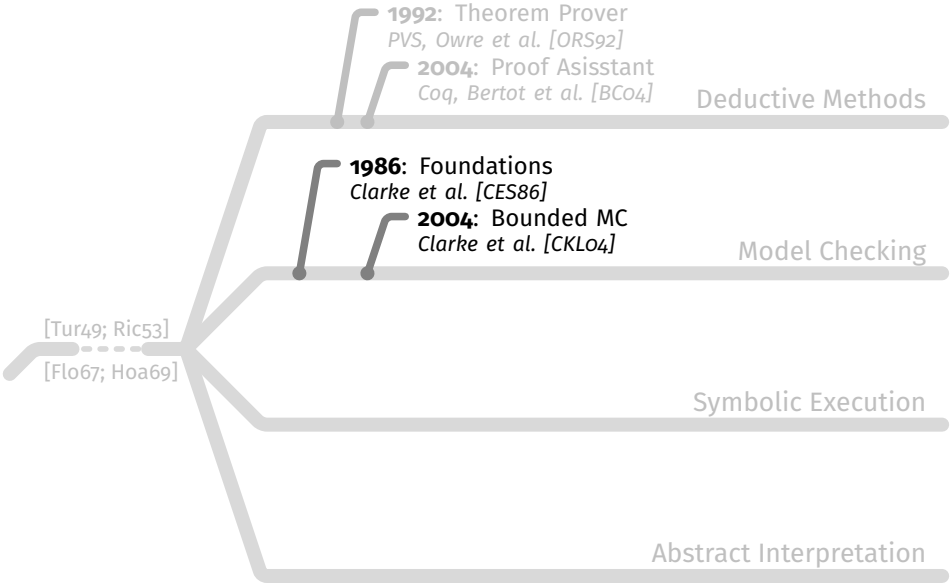

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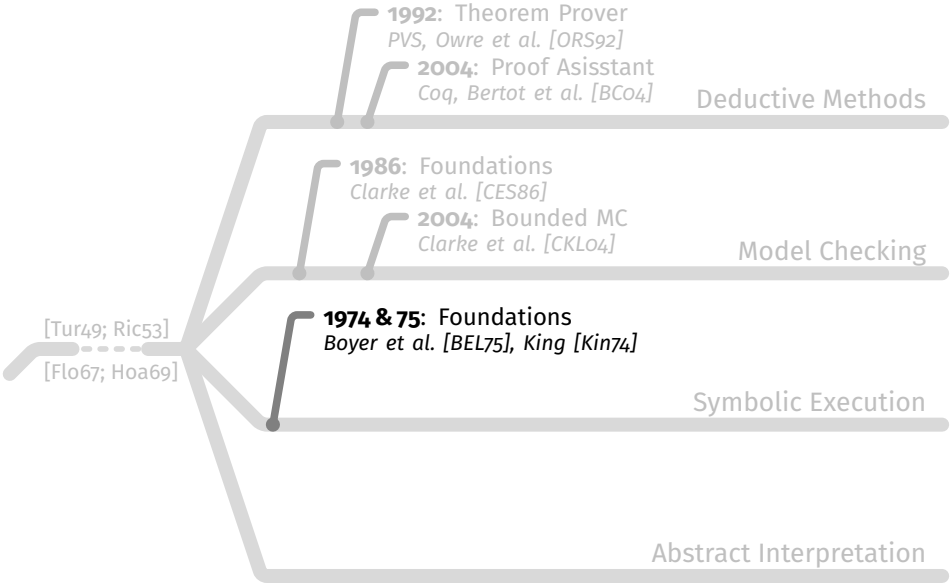

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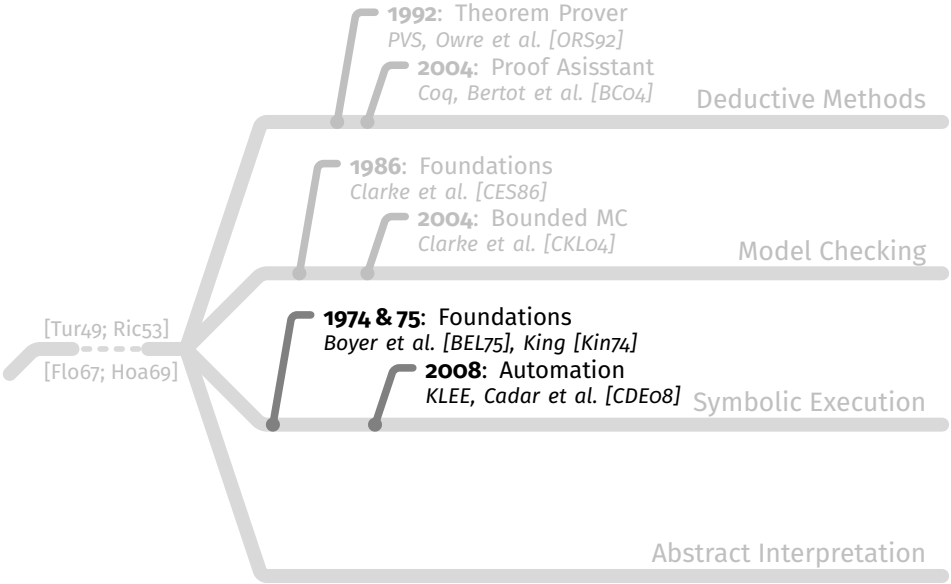

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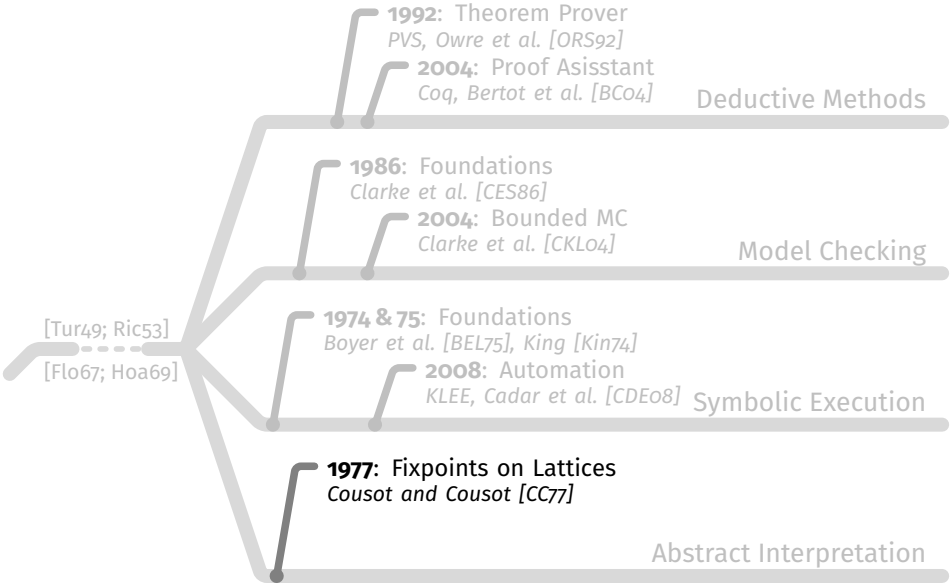

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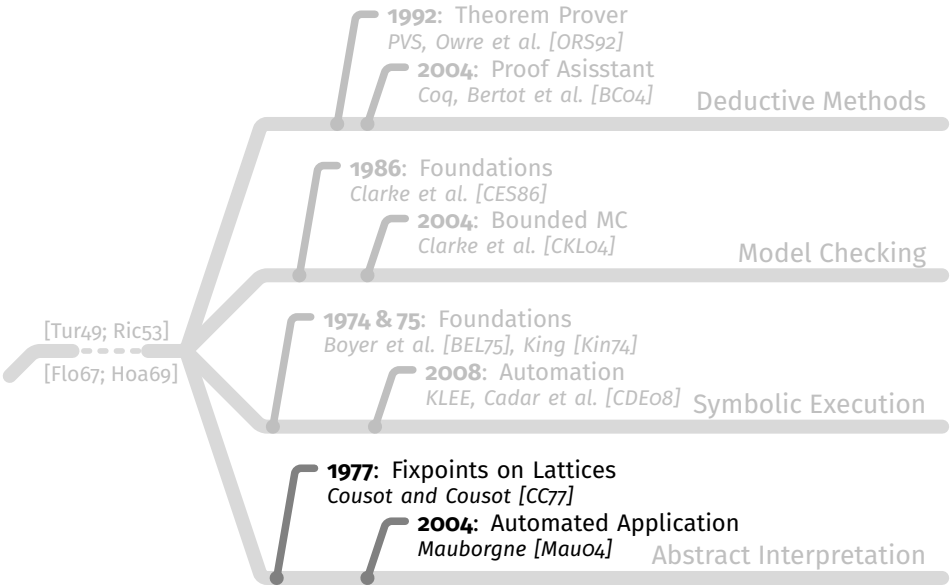

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